The role of emotional processing in metaphor comprehension: Evidence from ERPs in a cross-modal masked priming paradigm

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People often use metaphors to express emotion, as in “I am *burning* with anger,” or “The party *warmed* her spirits” [1]. Recent studies proposed that metaphors also enhance emotional processing [2,3]. Here we asked whether metaphor processing leads to more emotional processing, or whether emotional processing is integral and necessary for metaphor comprehension. Our main hypothesis was that emotional processing is available but not necessary for metaphoric processing. To test this, we used event-related potentials (ERPs) and a cross-modal masked priming paradigm, adapted from a metaphor study [4] that examined a different question, namely, whether a literal prime word could facilitate the processing of metaphoric sentences. While those researchers were concerned with a different question, we reasoned that the logic of their approach applies here: if emotional processing is indeed an integral part of metaphor processing, then an emotional prime word should facilitate the processing of a related metaphorical sentence more than a neutral prime word, when holding semantic relatedness constant between prime and target. For the dependent variables (ERP effects), we focused on N400 and late positivity (LP) [5].

So far, 30 native-speaker undergraduates (*M*age = 19.3), out of the planned 35 based on sample size calculation, participated in a cross-modal masked priming EEG experiment. Stimuli were 80 emotional metaphoric and 80 emotional literal sentences in English, matched for valence, arousal, familiarity and cloze probability. Each was paired with three types of primes: semantically related and emotional, semantically related and neutral, and semantically unrelated words (A). Prime words were also normed for several properties, as in (B). Modeling our procedure after [4], in each trial (C), participants listened to sentences containing target words while wearing an EEG cap. Shortly before the onset of the auditory target word, a masked prime was presented on a computer screen. To verify visual and auditory attention, each trial was followed by a color task and a probe task. For the color task, participants responded if the mask changed color via a button press (true for 40% of trials). In the probe task, participants responded if a word on the screen was present in the sentence they heard (true for 50% of trials).

Mean color task accuracy was .97 (*SD* = .03) and mean probe task accuracy was .96 (*SD* = .03), verifying good participation. ERPs of the metaphor-literal difference are displayed in (D). Semantically unrelated primes are visually more negative at N400 and earlier time window (150-400ms) than semantically related primes. A preliminary repeated measures ANOVA of 2 condition x 3 prime showed a marginal effect of prime type (p = .069). Planned pairwise comparisons suggested a marginally significant difference between the emotional prime and the unrelated prime (p = .065). Further, at the LP window (450-650), emotional primes are visually more positive compared to other prime conditions, not corroborated by preliminary analysis.

With our complete sample, we hope to verify that semantically related words indeed show a decreased N400, suggesting facilitated processing. Tentatively, if we do find that emotional primes decrease the N400 significantly and/or modulate later components at LP, this might mean that emotional processing a is more important to metaphor processing than previously thought. Our results will contribute to the field by further teasing apart the relationship between metaphor and emotion, while providing insights on the neural and pragmatic effects of metaphor use.

(A) Sample items, target words underscored

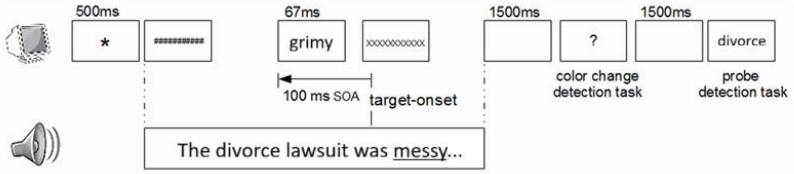
Condition	Sentence	Emotional prime	Neutral prime	Unrelated prime
Metaphor	Their divorce lawsuit was messy at the beginning.	grimy	scruffy	devout
Literal	The operating table was messy during surgery.	grimy	scruffy	devout
Metaphor	The country's stock market crashed while she worked.	wreck	impact	advise
Literal	The plane full of passengers crashed against the building.	wreck	impact	advise

(B) Statistical comparisons between prime conditions in *t*-values (*df*)

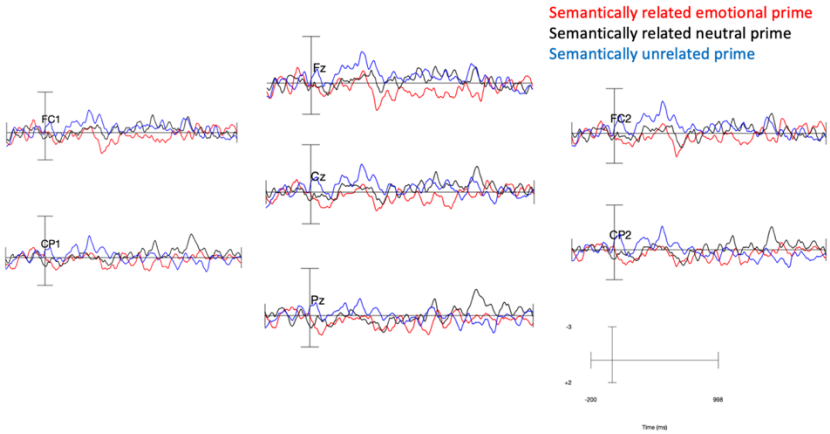
	Emotional vs. neutral	Neutral vs. unrelated	Unrelated vs. emotional
Arousal	**8.668 (158) *****	0.158 (158)	**9.264 (158) *****
Valence	**-10.218 (158) *****	0.122 (158)	**-10.204 (158) *****
Frequency	-1.004 (158)	0.983 (158)	-0.128 (158)
Number of Syllables	0.360 (156)	-0.381 (154)	0.025 (154)
Number of Phonemes	-0.042 (156)	0.173 (154)	0.119 (154)
Relatedness to sentence target	-0.053 (156)	**3.697 (156) *****	**5.091 (156) *****
Contextual relatedness	1.352 (318)	**15.702 (318) *****	**16.570 (318) *****

*** Significant differences at alpha < .001

(C) Procedure example



(D) ERP results (metaphor-literal difference waves) for each prime type at Fz, FC1, FC2, Cz, CP1, CP2, and Pz electrodes.



References: [1] Lakoff, G. & Johnson, M. (1980). [2] Citron, F. M., & Goldberg, A. E. (2014). [3] Mon, S. K., Nencheva, M., Citron, F. M., Lew-Williams, C., & Goldberg, A. E. (2021). [4] Weiland, H., Bambini, V., & Schumacher, P. (2014). [5] Baiocco, L., Kielar, A., & Lai, V. T. (2026).